

# **AD162S Datasheet**

**Zhuhai Jieli Technology Co.,LTD**

**Version: 1.0**

**Date: 2023.08.14**

## AD162S Features

### CPU

- 32bit DSP
- Maximum speed 160MHz
- Interrupts with 8 priority level

### Memory

- OTP
- Optional built-in flash memory

### Clocks

- On-chip 16 MHz clock
- On-chip 200KHz lower-temperature-drift clock

### DSP Audio Processing

- Support MP2, MP3, WMA, WAV decoding
- Multi-band DRC limiter
- Multi-band EQ configuration for voice Effects

### Audio Codec

- One channels 16-bit DAC, with SNR  $\geq$  93dB
- One channel 24-bit ADC, SNR  $\geq$  85dB
- Audio DAC Sampling rates of 8KHz/11.025KHz/16KHz/22.05KHz/24KHz/32KHz/44.1KHz/48KHz/64KHz/88.2KHz/96KHz are supported
- Audio ADC Sampling rates of 8KHz/11.025KHz/16KHz/22.05KHz/24KHz/32KHz/44.1KHz/48KHz are supported
- Audio DAC support single-ended mode
- Support analog audio input
- Support for driving 16 or 32 ohm speaker

### Peripherals

- One full speed USB OTG controller
- One SD host controller for MMC/SD
- Three multi-function 32-bit timers, support capture and PWM mode
- UART0 controller
- The UART1 supports DMA and flow control
- One IIC Master controller
- Two SPI Master / Slaver controller with DMA
- One QDEC interface
- 8-channel 10-bit general purpose ADC
- 4-channel Advance PWM controller
- 11 Individually programmable and multiplexed GPIO pins
- Digital peripheral crossbar
- Up to 11 external interrupt / wake-up source ( low power available, can be multiplexed to any I/O )

### PMU

- Built-in lithium battery charging manager, up to 120mA charging current
- Less than 2uA soft off current
- VPWR range : 4.5V to 6.0V
- VBAT range : 2.2V to 5.0V
- IOVDD range : 2.1V to 3.6V

### Packages

- SOP16

### Temperature

- Operating temperature: -40°C to +85°C
- Storage temperature: -65°C to +150°C

### Applications

- Audio player
- Microcontrollers

## 1 Block Diagram

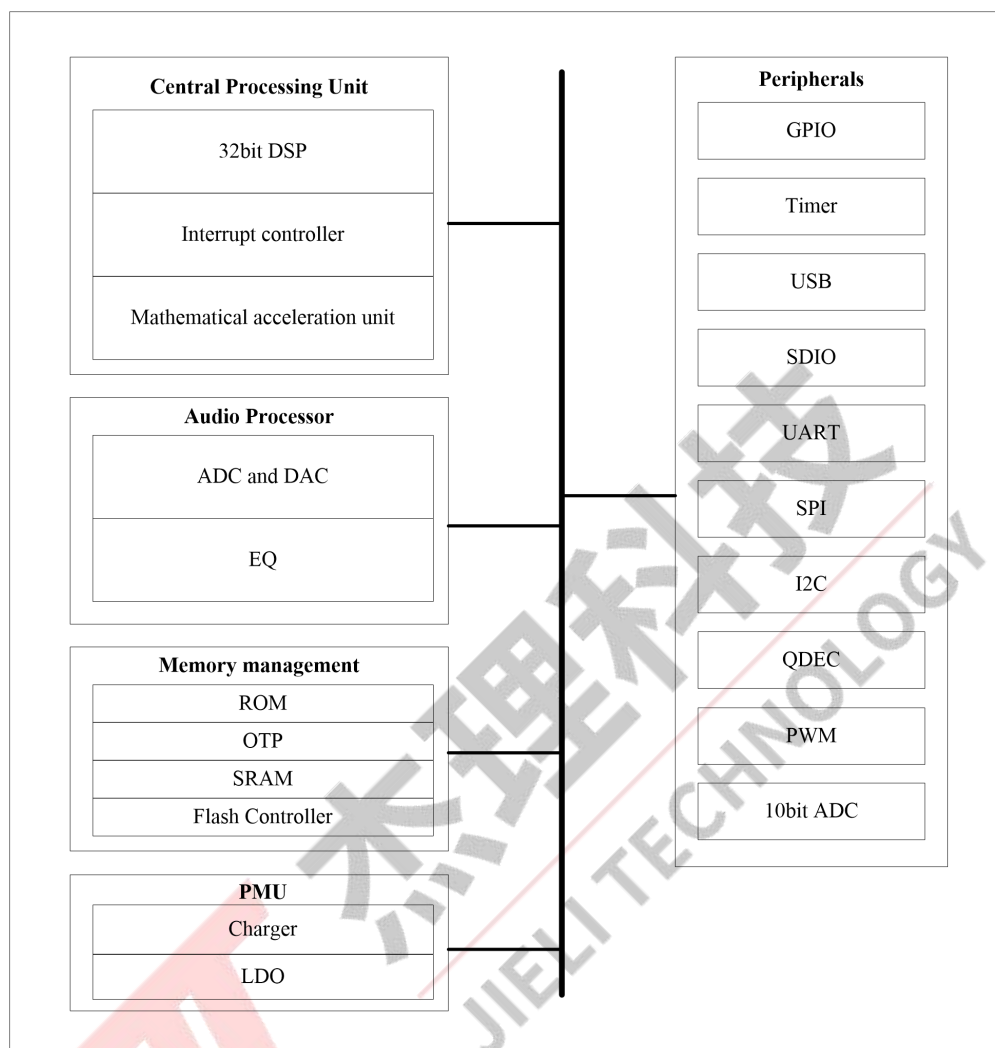


Figure 1-1 AD162S Block Diagram

## 2 Pin Definition

### 2.1 Pin Assignment

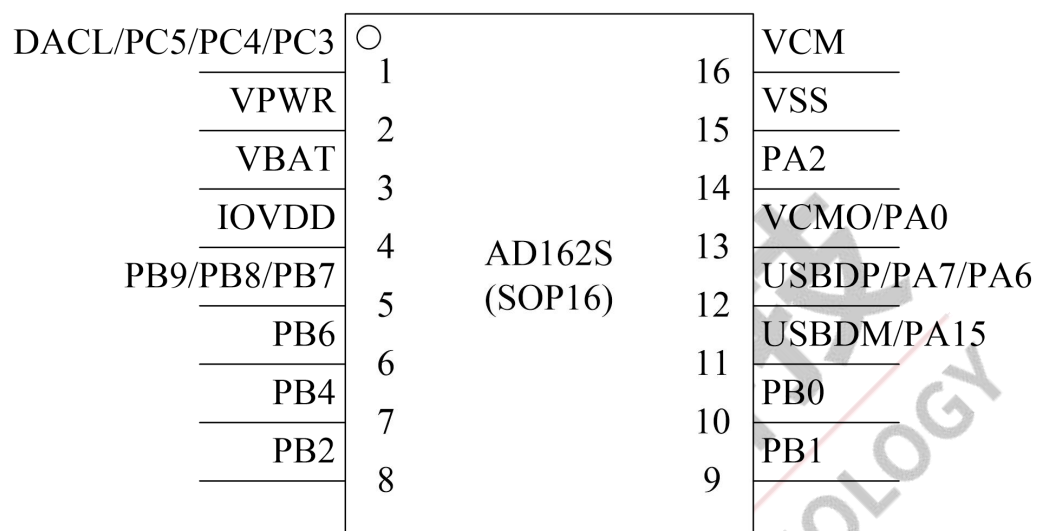


Figure 2-1 AD162S Package Diagram

## 2.2 Pin Description

**Table 2-1 AD162S Pin Description**

| PIN NO. | Name       | Type     | Function                      | Other Function                                                                                                                                  |
|---------|------------|----------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | DACL       | AO       |                               | Left channel audio output;                                                                                                                      |
|         | PC5        | I/O      | GPIO                          | AINR:Right channel analog audio input;                                                                                                          |
|         | PC4        | I/O      | GPIO                          | AINL:Left channel analog audio input;<br>TMR2:Timer2 Clock Input;                                                                               |
|         | PC3        | I/O      | GPIO                          | SDPG:SD card Power Gate;<br>ADC13:ADC Input Channel 13;                                                                                         |
| 2       | VPWR (PP0) | PI (I/O) | GPIO                          | Charge Power Input;<br>UART1TXA:Uart1 Data Output(A);<br>UART1RXA:Uart1 Data Input(A);<br>CAP1:Timer1 Capture;                                  |
| 3       | VBAT       | P        |                               | Battery interface;                                                                                                                              |
| 4       | IOVDD      | PO       | Power supply for GPIO         | Built-in linear voltage regulator output;                                                                                                       |
| 5       | PB9        | I/O      | GPIO                          | CLKOUT2:Clock Out2;                                                                                                                             |
|         | PB8        | I/O      | GPIO                          | ADC10:ADC Input Channel 10;<br>CLKOUT1:Clock Out1;                                                                                              |
|         | PB7        | I/O      | GPIO                          | Q-decoder_0;<br>ADC9:ADC Input Channel 9;<br>CLKOUT0:Clock Out0;                                                                                |
| 6       | PB6        | I/O      | GPIO                          | ADC8:ADC Input Channel 8;                                                                                                                       |
| 7       | PB4        | I/O      | GPIO                          | SPI1DI(B):SPI1 Data In(B);<br>SDDAT(A):SD Data(A);<br>ADC7:ADC Input Channel 7;                                                                 |
| 8       | PB2        | I/O      | GPIO                          | SPI1CLK(B):SPI1 Clk(B);<br>SDCLK(A/B):SD Clock(A/B);<br>ADC6:ADC Input Channel 6;                                                               |
| 9       | PB1        | I/O      | GPIO                          | SPI1DI(A):SPI1 Data In(A);                                                                                                                      |
| 10      | PB0        | I/O      | GPIO                          | ADC5:ADC Input Channel 5;                                                                                                                       |
| 11      | UDBDM      | I/O      | USB Negative Data (pull down) | SDDAT(B):SD Data(B);<br>SPI1DO(A):SPI1 Data Out(A);<br>IIC0_SDA(A):IIC0 SDA(A);<br>UART0RXA:Uart0 Data Input(A);<br>ADC15:ADC Input Channel 15; |
|         | PA15       | I/O      | GPIO                          | PWM1:Timer1 PWM Output;                                                                                                                         |
| 12      | USBDP      | I/O      | USB Positive Data (pull down) | SPI1CLKA:SPI1 Clk(A);<br>IIC0_SCL(A):IIC0 SCL(A);<br>UART0TXA:Uart0 Data Output(A);<br>ADC14:ADC Input Channel 14;                              |
|         | PA7        | I/O      | GPIO                          |                                                                                                                                                 |

|    |      |     |      |                                                                                                                                              |
|----|------|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------|
|    | PA6  | I/O | GPIO | ADC2:ADC Input Channel 2;                                                                                                                    |
| 13 | PA0  | I/O | GPIO | MICLDO:Microphone linear voltage regulator output;<br>ADC0:ADC Input Channel 0;<br>UART1TXB:Uart1 Data Output(B);<br>PWM0:Timer0 PWM Output; |
|    | VCMO | AO  |      | Negative of earphone;                                                                                                                        |
| 14 | PA2  | I/O | GPIO | MICIN1:MIC1 Input Channe;<br>UART1_CTS:Uart1 clear to send;<br>PWMCH0H:Motor PWM Channel0(H);                                                |
| 15 | VSS  | G   |      | System ground;                                                                                                                               |
| 16 | VCM  | P   |      | Audio analog reference bias;                                                                                                                 |

| Pin Type | Description   | Pin Type | Description     |
|----------|---------------|----------|-----------------|
| P        | Power         | I/O      | Input or Output |
| PI       | Power Input   | I        | Input           |
| PO       | Power Output  | O        | Output          |
| AO       | Analog Output | G        | Ground          |

## 3 Electrical Characteristics

### 3.1 Absolute Maximum Ratings

Table 3-1

| Symbol             | Parameter                | Min  | Max       | Unit |
|--------------------|--------------------------|------|-----------|------|
| T <sub>opt</sub>   | Operating temperature    | -40  | +85       | °C   |
| T <sub>stg</sub>   | Storage temperature      | -65  | +150      | °C   |
| V <sub>BAT</sub>   | Supply Voltage           | -0.3 | 5.0       | V    |
| V <sub>PWR</sub>   | Charger Voltage          | -0.3 | 6.0       | V    |
| V <sub>IOVDD</sub> | Voltage applied at IOVDD | -0.3 | 3.6       | V    |
| V <sub>GPIO</sub>  | Voltage applied to GPIO  | -0.3 | IOVDD+0.3 | V    |

Note : The chip can be damaged by any stress in excess of the absolute maximum ratings listed below

### 3.2 PMU Characteristics

Table 3-2

| Symbol           | Parameter              | Min | Typ | Max | Unit | Test Conditions                       |
|------------------|------------------------|-----|-----|-----|------|---------------------------------------|
| V <sub>BAT</sub> | Voltage Input          | 2.2 | 3.7 | 5.0 | V    | —                                     |
| V <sub>PWR</sub> | Charger supply Voltage | 4.5 | 5.0 | 6.0 | V    | —                                     |
| IOVDD            | Voltage output         | 2.1 | 3.0 | 3.6 | V    | V <sub>BAT</sub> = 4.2V, 10mA loading |
|                  | Loading current        | —   | —   | 100 | mA   | IOVDD=3.3V@V <sub>BAT</sub> = 3.6V    |
| V <sub>LVD</sub> | Voltage input          | 2.1 | 2.8 | 2.8 | V    | Low-Voltage Detection of IOVDD        |

### 3.3 Battery Charge

Table 3-3

| Symbol                 | Parameter                            | Min  | Typ  | Max  | Unit | Test Conditions                               |
|------------------------|--------------------------------------|------|------|------|------|-----------------------------------------------|
| V <sub>PWR</sub>       | Charge Input Voltage Range           | 4.5  | 5    | 6.0  | V    | —                                             |
| V <sub>BAT Float</sub> | Battery Charge Termination Voltage   | 4.15 | 4.2  | 4.25 | V    | V <sub>PWR</sub> > 4.5V                       |
|                        |                                      | 4.30 | 4.35 | 4.40 | V    | V <sub>PWR</sub> > 4.65V                      |
| I <sub>BAT</sub>       | Fast Charge Current                  | 20   | —    | 120  | mA   | V <sub>BAT</sub> =4.0V@V <sub>PWR</sub> =5.0V |
| I <sub>END</sub>       | Charge Termination Current Threshold | 2    | —    | 12   | mA   | CHG_IIFULL_S=0,1                              |
| V <sub>Trikl</sub>     | Trickle Charge Voltage               | —    | 3.0  | —    | V    | V <sub>PWR</sub> > 4.5V                       |
| I <sub>Trikl</sub>     | Trickle Charge Current               | 2    | —    | 12   | mA   | V <sub>BAT</sub> < V <sub>Trikl</sub>         |

### 3.4 IO Input/Output Electrical Logical Characteristics

Table 3-4

| GPIO input characteristics  |                           |            |     |            |      |                 |
|-----------------------------|---------------------------|------------|-----|------------|------|-----------------|
| Symbol                      | Parameter                 | Min        | Typ | Max        | Unit | Test Conditions |
| $V_{IL}$                    | Low-Level Input Voltage   | -0.3       | —   | 0.3* IOVDD | V    | IOVDD = 3.0V    |
| $V_{IH}$                    | High-Level Input Voltage  | 0.7* IOVDD | —   | IOVDD+0.3  | V    | IOVDD = 3.0V    |
| GPIO output characteristics |                           |            |     |            |      |                 |
| Symbol                      | Parameter                 | Min        | Typ | Max        | Unit | Test Conditions |
| $V_{OL}$                    | Low-Level Output Voltage  | —          | —   | 0.1* IOVDD | V    | IOVDD = 3.0V    |
| $V_{OH}$                    | High-Level Output Voltage | 0.9* IOVDD | —   | —          | V    | IOVDD = 3.0V    |

### 3.5 Internal Resistor Characteristics

Table 3-5

| Port                                                         | Drive Current                                                                        | Internal Pull-Up Resistor | Internal Pull-Down Resistor | Comment                                                                                            |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------|-----------------------------|----------------------------------------------------------------------------------------------------|
| PA0,PA2<br>PA6,PA7,PA15<br>PB0~PB2,PB4<br>PB6~PB9<br>PC3~PC5 | 2mA(HD1,HD0==0,0)<br>5.6mA(HD1,HD0==0,1)<br>18mA(HD1,HD0==1,0)<br>30mA(HD1,HD0==1,1) | 10K                       | 10K                         | 1. USBDM,USBDP default pull down<br>2. Internal pull-up/pull-down resistance   accuracy $\pm 20\%$ |
| PP0(VPWR)                                                    | 1.4mA                                                                                | 10K                       | 10K                         |                                                                                                    |
| USBDP                                                        | 27mA                                                                                 | 1.5K                      | 15K                         |                                                                                                    |
| USBDM                                                        |                                                                                      | 180K                      | 15K                         |                                                                                                    |

### 3.6 Audio DAC Characteristics

Table 3-6

| Parameter          | MODE         | Min | Typ | Max | Unit  | Test Conditions                                                     |
|--------------------|--------------|-----|-----|-----|-------|---------------------------------------------------------------------|
| Frequency Response |              | 20  | —   | 20K | Hz    | 1KHz/0dB<br>10k ohm loading<br>With A-Weighted Filter<br>IOVDD>2.7V |
| Output Swing       | Single-ended | —   | 750 | —   | mVrms |                                                                     |
| THD+N              | Single-ended | —   | -80 | —   | dB    |                                                                     |
| S/N                | Single-ended | —   | 93  | —   | dB    |                                                                     |
| Dynamic Range      | Single-ended | —   | 93  | —   | dB    |                                                                     |
| Noise Floor        | Single-ended | —   | 18  | —   | uVrms |                                                                     |
| Crosstalk          | Single-ended | —   | -93 | —   | dB    | 10KHz/0dB<br>10k ohm loading<br>IOVDD>2.7V                          |

### 3.7 Audio ADC Characteristics

Table 3-7

| Parameter     | MODE         | Min | Typ | Max | Unit | Test Conditions                                                                    |
|---------------|--------------|-----|-----|-----|------|------------------------------------------------------------------------------------|
| Dynamic Range | Single-ended | —   | 85  | —   | dB   | Fsample=44.1KHz,Gain=-2dB<br>Fin=1KHz @1Vpp<br>NO A-wt 20Hz-20KHz<br>IOVDD>2.7V    |
|               |              | —   | 72  | —   | dB   | Fsample=44.1KHz,Gain=14dB<br>Fin=1KHz @160mVpp<br>NO A-wt 20Hz-20KHz<br>IOVDD>2.7V |
| S/N           | Single-ended | —   | 85  | —   | dB   | Fsample=44.1KHz,Gain=-2dB<br>Fin=1KHz @1Vpp<br>NO A-wt 20Hz-20KHz<br>IOVDD>2.7V    |
|               |              | —   | 72  | —   | dB   | Fsample=44.1KHz,Gain=14dB<br>Fin=1KHz @160mVpp<br>NO A-wt 20Hz-20KHz<br>IOVDD>2.7V |
| THD+N         | Single-ended | —   | -78 | —   | dB   | Fsample=44.1KHz,Gain=-2dB<br>Fin=1KHz @1Vpp<br>NO A-wt 20Hz-20KHz<br>IOVDD>2.7V    |
|               |              | —   | -70 | —   | dB   | Fsample=44.1KHz,Gain=14dB<br>Fin=1KHz @160mVpp<br>NO A-wt 20Hz-20KHz<br>IOVDD>2.7V |

## 4 Package Information

### 4.1 SOP16

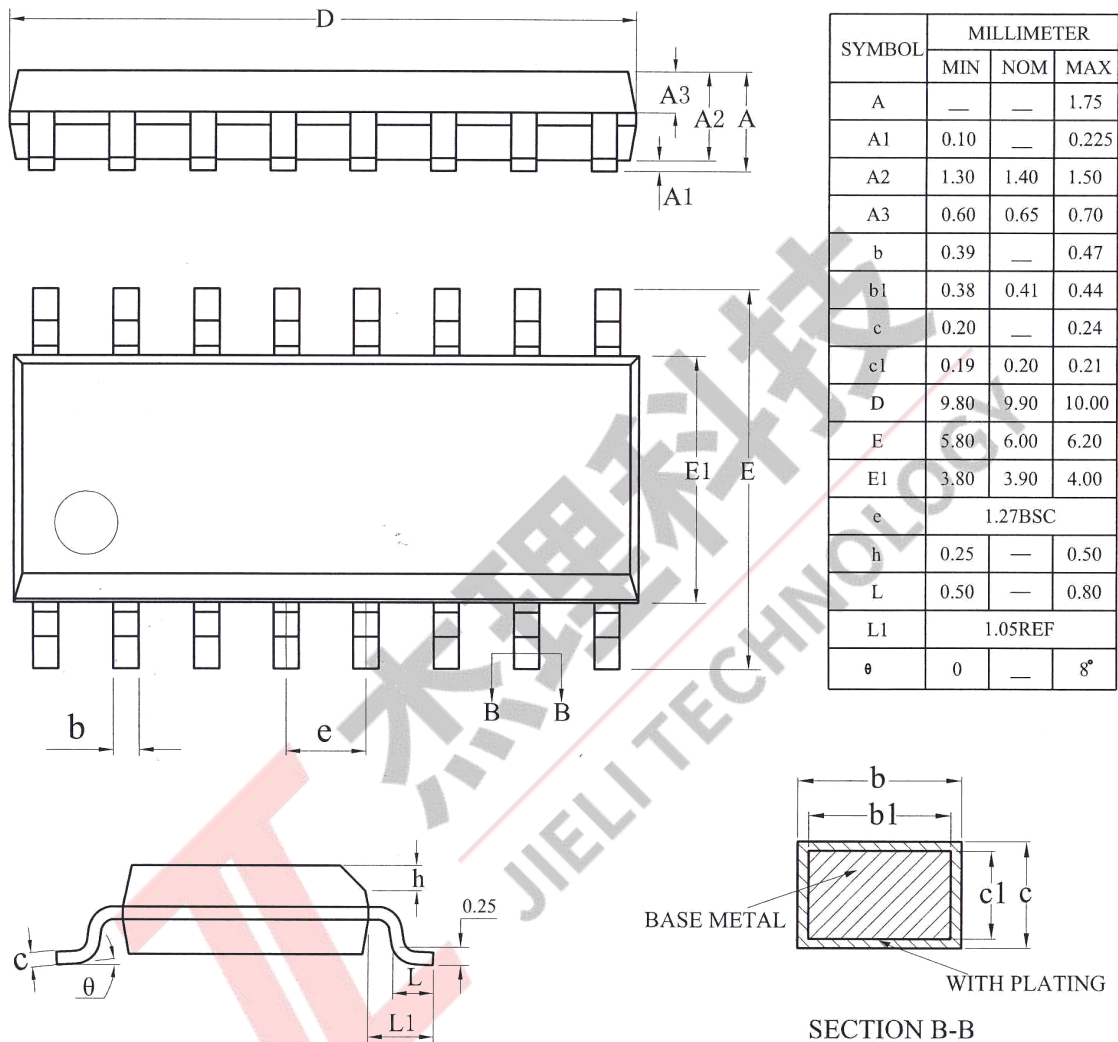
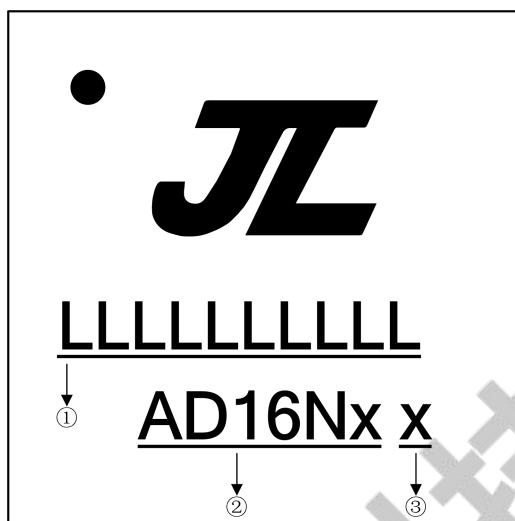


Figure 4-1 AD162S Package

## 5 IC Marking Information



① LLLLLLLLLL : Production Batch

② AD16Nx : Chip Model

③ Built-in flash size

0: No Flash Memory

2: 2Mbit Flash

4: 4Mbit Flash

8: 8Mbit Flash

6: 16Mbit Flash

3: 32Mbit Flash

## 6 Solder-Reflow Condition

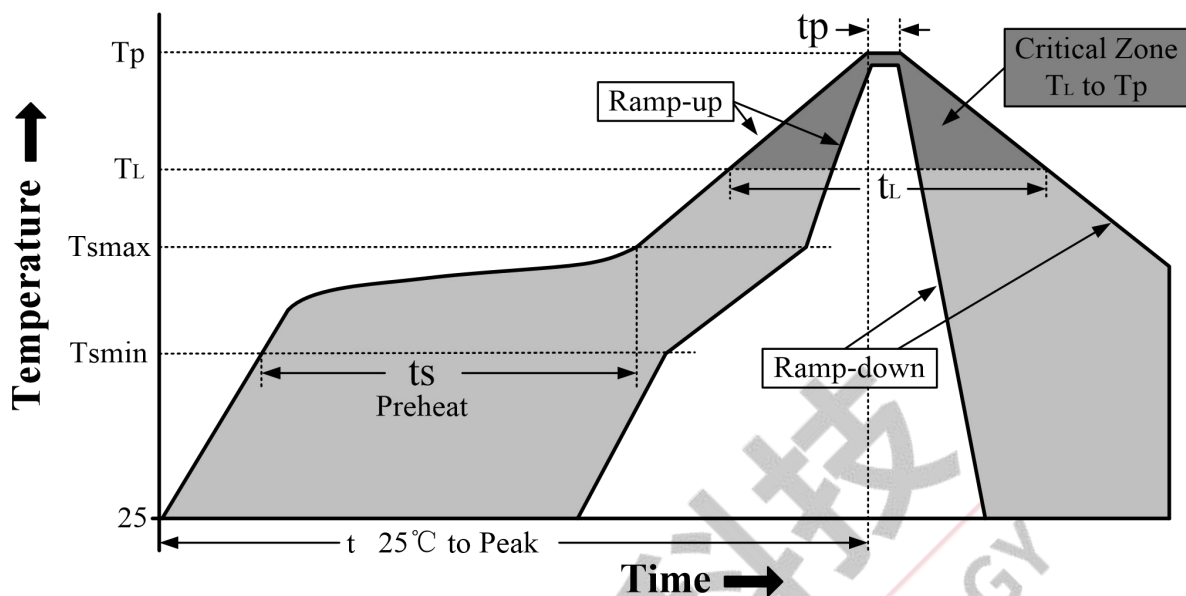


Figure 6-1 Classification Reflow Profile

### Classification Profiles

Table 6-1

| Profile Feature                                      |                                                  | Sn-Pb Eutectic Assembly | Pb-Free Assembly |
|------------------------------------------------------|--------------------------------------------------|-------------------------|------------------|
| Preheat/<br>Soak                                     | Temperature Min ( $T_{smin}$ )                   | 100 °C                  | 150 °C           |
|                                                      | Temperature Max ( $T_{smax}$ )                   | 150 °C                  | 200 °C           |
|                                                      | Time ( $t_s$ ) from ( $T_{smin}$ to $T_{smax}$ ) | 60-120 seconds          | 60-180 seconds   |
| Average ramp-up rate ( $T_{smax}$ to $T_p$ )         |                                                  | 3 °C/second max         | 3 °C/second max  |
| Liquidous temperature ( $T_L$ )                      |                                                  | 183 °C                  | 217 °C           |
| Time ( $t_L$ ) maintained above $T_L$                |                                                  | 60-150 seconds          | 60-150 seconds   |
| Peak package body temperature ( $T_p$ )              |                                                  | See Table 6-2.          | See Table 6-3.   |
| Time within 5°C of actual Peak Temperature ( $t_p$ ) |                                                  | 10-30 seconds           | 20-40 seconds    |
| Ramp-down rate ( $T_p$ to $T_L$ )                    |                                                  | 6 °C/second max.        | 6 °C/second max. |
| Time 25 °C to peak temperature                       |                                                  | 6 minutes max.          | 8 minutes max.   |

Note 1: All temperatures refer to topside of the package, measured on the package body surface.

Note 2: Time within 5°C of actual peak temperature ( $t_p$ ) specified for the reflow profiles is a “supplier” minimum and “user” maximum.

### SnPb - Classification Temperature

Table 6-2

| Package Thickness | Volume mm <sup>3</sup><br>< 350 | Volume mm <sup>3</sup><br>≥ 350 |
|-------------------|---------------------------------|---------------------------------|
| <2.5 mm           | 240 +0/-5 °C                    | 225 +0/-5 °C                    |
| ≥ 2.5 mm          | 225 +0/-5 °C                    | 225 +0/-5 °C                    |

**Pb-free - Classification Temperature**      **Table 6-3**

| Package Thickness | Volume mm <sup>3</sup><br>< 350 | Volume mm <sup>3</sup><br>350 - 2000 | Volume mm <sup>3</sup><br>> 2000 |
|-------------------|---------------------------------|--------------------------------------|----------------------------------|
| < 1.6mm           | 260 °C                          | 260 °C                               | 260 °C                           |
| 1.6 mm - 2.5mm    | 260 °C                          | 250 °C                               | 245 °C                           |
| > 2.5mm           | 250 °C                          | 245 °C                               | 245 °C                           |

## 7 Revision History

| Date       | Revision | Description      |
|------------|----------|------------------|
| 2023.08.14 | V1.0     | Initial Release. |
|            |          |                  |
|            |          |                  |
|            |          |                  |

